

Brief: WhatsApp Notification Migration

Status: Pre-Implementation / Pending Approval

Project: Transitioning Loan Communications from Email to WhatsApp

Executive Summary

Following the client request to migrate loan notifications from email to WhatsApp, I have completed the research phase. I recommend a **Queued Notification Service**, to ensure system resilience and auditability.

1. Research & Analysis

1.1 Scope of Review

- **Existing Infrastructure:** Evaluated current email-based verification and loan lifecycle triggers.
- **Technical Stack:** Assessed integration compatibility with the current backend and deployment pipeline.
- **Operations:** Identified requirements for reliability, retries, and failure observability.

1.2 Key Findings

- **Trigger Points:** The loan lifecycle (Application → Approval → Disbursement → Repayment → Completion) provides logical hooks for automated messaging.
- **Architectural Need:** Sending messages synchronously within route handlers poses a latency risk. An **Asynchronous Worker/Queue** model is necessary.
- **Compliance:** Strict adherence to **E.164 phone normalization** and explicit user consent is required.

1.3 Architectural Recommendation

Feature	Option A: Direct Integration	Option B: Queued Service (Recommended)
Complexity	Low (Initial)	Medium
Resilience	Poor (No retries)	High (Automatic retries)
Observability	Minimal	Comprehensive (Audit logs)
Scalability	Limited	High

2. Proposed Implementation Plan

2.1 Technical Approach

1. **Abstraction:** Create a generic notification layer to handle multiple providers.
2. **Queueing:** Implement a database-backed outbox or message queue.

3. **Tracking:** Use webhooks to monitor delivery status (*Sent, Delivered, Read, Failed*).
4. **Safeguards:** Deploy feature flags to allow instant rollback to email if needed.

2.2 Phased Roadmap

- **Phase 1: Foundation:** Schema design, queue setup, and worker initialization.
- **Phase 2: Integration:** Provider API setup (Send/Receive) and webhook handling.
- **Phase 3: Wiring:** Connecting loan lifecycle events to the new service.
- **Phase 4: Admin UI:** Visibility into delivery logs and user consent status.
- **Phase 5: Pilot:** Controlled rollout to a limited user group.

2.3 Risk Mitigation

- **Template Rejection:** Submit templates for approval during Phase 1.
- **Data Integrity:** Enforce strict E.164 validation at the point of entry.
- **Race Conditions:** Implement **idempotency keys (unique, client-generated identifiers)** per event to prevent duplicate messages.

3. Documentation Reference

Document	Purpose	Location
Technical Architecture	Detailed system design and data flow	WHATSAPP_COMMUNICATION_ARCHITECTURE.pdf
Execution Plan	Granular task breakdown and timeline	WHATSAPP_MIGRATION_PLAN.pdf

4. Immediate Action Items (Approval Required)

1. **Provider Selection:** Meta Cloud API vs. Twilio.
2. **Templates:** Approval of initial message scripts.
3. **Consent:** UX placement and wording for user opt-in.
4. **Pilot Scope:** Determination of the initial test group.

5. Proposed Internal Update

I have finalized the pre-implementation brief for the WhatsApp notification migration. The research favors a queued architecture to ensure reliability.